CSCI160 Computer Science I
Lab Exercise

Objectives

- Understand and evaluate boolean expressions
- Implement conditional logic using if statements
- Develop a voting eligibility checker
- Work with dictionaries and lists
- Debug and test code with test cases

AI Statement: You are allowed to interact with an AI tool during this assignment. If you do this you are ***required*** to cite this usage in your submission file. Failure to do so will be considered a violation of academic integrity and penalized.

Given the following boolean expressions, determine the result for the given values of `A`, `B`, and `C`. Choose the correct answer and provide an explanation.

Answer these questions in a document. You will be submitting this document as a PDF when finished.

Check your conclusions by running these as programs. Step through with the debugger to track the execution path.

Example Evaluation:

`A = True, B = False, C = True`

1. `(A AND B) OR C = True`

   `True (A) AND False (B) is` **`False`**
   `False (from above) OR True (C) is` **`True`**

   `The entire expression is True`

2. `B OR NOT A = False`

   `NOT True (A) evaluates first and is` **`False`**
   `False (B) OR False (from above) is` **`False`**

   `The entire expression is False`

1. Evaluate Simple Boolean Expressions

`A = True, B = False, C = True`

1. `A AND B`
2. `A OR B`
3. `NOT A`
4. `A AND (B OR C)`

2. Evaluate Nested Boolean Expressions:

`A = False, B = True, C = False`

1. `(A OR B) AND (NOT C)`
2. `NOT (A AND B) OR C`
3. `(A OR NOT B) AND (B OR C)`

3. Boolean Logic with Comparisons

`A = 5, B = 10, C = 15`

1. `(A < B) AND (B < C)`
2. `(A == B) OR (B != C)`
3. `NOT (A > B) AND (C >= B)`

4. Boolean Logic in Conditional Statements

Given the following Python code snippet, determine the output for the given values of A, B, and C:

```
A = True
B = False
C = True

if (A and B) or (not C):
    print("Case 1")
elif (A or B) and (not C):
    print("Case 2")
else:
    print("Case 3")
```

Python Coding Exercise: Voting Eligibility Checker

Objective:
Write a program that checks whether a person is eligible to vote. If they are deemed eligible, the program will add them to the database of voters (Python Dictionary), show the updated database and display their representative.

Voting Criteria:

1. **Age**: Must be **18 years or older**.
2. **Citizenship**: Must be a **citizen** of the country.
3. **District**: Must be a **resident of a NY voting district.**
4. **Registered**: Must be **registered to vote**.

Problem Breakdown

1. **Input Collection**

    - Using clear messages, prompt the user to enter their:
        - **Name**
        - **Age** (as a number)
        - **Citizenship** (yes/no)
        - **District** (as a number)
        - **Registration** status (yes/no)

2. **Eligibility Check using *inclusive logic***

    - Use **AND** logic to enforce that ***every condition*** must be true for eligibility. If any one of the inputs does not meet the listed criteria, the person is not eligible to vote.
    - Use an `if` statement with the `and` operator to check if **all conditions** are true. They must be over 18 and a citizen and in a valid NY district and registered to vote
    - If they are eligible to vote, do the following
        - print a **success message**: `"You are eligible to vote!"`
        - add them to the voting database (Python dictionary)
        - display their representative and their political party
        - display the updated database and the number of voters in the database. You can get the keys from a dictionary by using the `keys()` method

3. **Detailed Failure Feedback**

    - If **any** of the 4 conditions are false, print: `"You are NOT eligible to vote."`
    - Then provide a reason for **each unmet condition**. Here is an example

Enter your age: 16
Are you a citizen? (yes/no): no
Enter your voting district: 19
Are you registered to vote? (yes/no): no

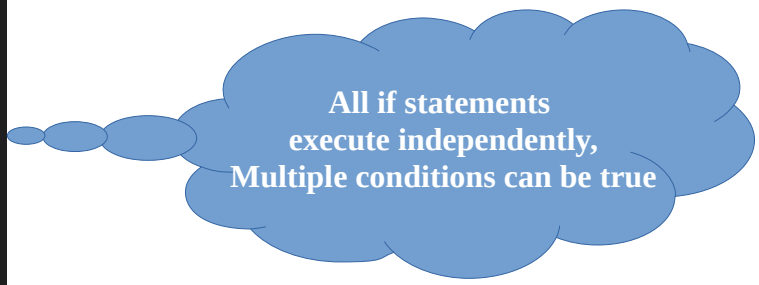
You are NOT eligible to vote.

You must be at least 18 years old to vote.
You must be a citizen to vote.
You must be registered to vote.

4. **Independent Evaluation:** Use the concept of ***independent evaluation*** to determine which failure message to display. Independent evaluation exists when you define a collection of separate `if` statements rather than an `if-else if` chain. This type of logic behaves differently in terms of execution flow. Here's an example of the two approaches.

Independent Evaluation using separate if statements

```
num = 10
if num > 5:
    print("Greater than 5")
if num < 15:
    print("Less than 15")
if num < 20:
    print("Less than 20")
```



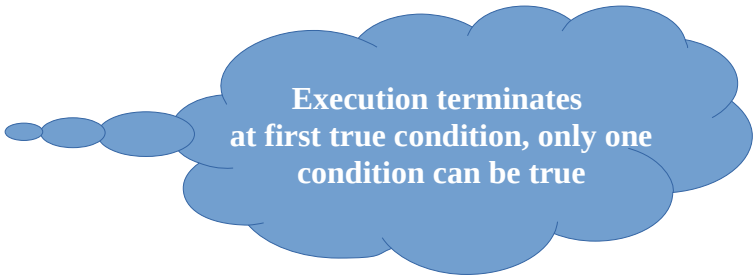
Produces

Greater than 5
Less than 15
Less than 20

Each `if` statement is evaluated independently: Multiple `if` statements can be true at the same time, leading to multiple blocks of code executing.

Exclusive Evaluation using single if/else if block

```
num = 10
if num > 5:
    print("Greater than 5")
elif num < 15:
    print("Less than 15")
elif num < 20:
    print("Less than 20")
```



Produces: Greater than 5

Only one condition can be true: Once a condition is met, no further conditions are checked and the if block is exited with execution continuing at the next un-indented code

Handling the user input

- Allow the user to enter input using any mixture of upper/lower case letters
  - Yes, yes, yEs, YES . . . etc
  - Use one of the built-in String methods **`upper()`** or **`lower()`** to force the input to either upper case or lower case and then perform the comparison.
  - **Example:** `"yes".upper()` will return `"YES"`
  - **Example:** `"YES".lower()` will return `"yes"`
- For the user's name use the `capitalize()` method to force the input to standard casing
  - **Example:** `"sPoNgEbOb".capitalize()` will return `"Spongebob"`
  - **Challenge:** Ensure both the first and last name are capitalized correctly. Check out the `split()` method

```
>>> name = "Spongebob Squarepants"
>>> name.split()
['Spongebob', 'Squarepants']
```

Download the Python file named: `voters.py`

- This file has a bit of starter code including two data structures that you ***must use***
  - Dictionary of NY voting districts and their representative
  - Dictionary of voters
- Add your authorship header comment block to the top of the file. Include a brief description of the problem
- Write Python code to implement a solution to this problem as defined.

- **Documentation:** You must include descriptive comments and AI citations for all AI generated code that you include.

Test cases

```
Enter your name: sPonGEbOB squAREpaNTs
Enter your age: 21
Are you a U.S. citizen? (yes/no): yES
Enter your district number: 19
Are you registered to vote? (yes/no): YeS

Spongebob Squarepants, you are eligible to vote!

Your representative is Josh Riley from the Democrat party.

Voters:  ['Frank N. Stein', 'Jane Jetson', 'Jim Jeffries', 'Jill Jones', 'Spongebob Squarepants']

There are 5 voters.
```

```
Enter your name: Patrick Star
Enter your age: 16
Are you a U.S. citizen? (yes/no): no
Enter your district number: 23
Are you registered to vote? (yes/no): no

You are not eligible to vote because you are under 18 years old.
You are not eligible to vote because you are not a U.S. citizen.
You are not eligible to vote because you are not registered to vote.
```

```
Enter your name: Squid Ward
Enter your age: 47
Are you a U.S. citizen? (yes/no): yes
Enter your district number: 37
Are you registered to vote? (yes/no): yes

You are not eligible to vote because you are not in a valid district.
```

Submission: When finished attach your modified ***voters.py*** and the PDF document from part one of the lab.